

Lunar Entertainment Vehicle I Proposal

Concept of Operations

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Acronyms

CG	Center of Gravity. 11
ECC	Error Correcting Code. 31
ECLSS	Environmental Control and Life Support System. 12, 28, 30–32, 34
EVA	Extravehicular activity. 9, 24
GNC	Guidance Navigation and Control. 12
HLS	Human Landing System. 3, 27
IDSS	International Docking System Standard. 3, 18
ISRU	In-Situ Resource Utilization. 25
ISS	International Space Station. 18, 19
IWS	In Wheel Suspension. 16
LEO	Low Earth Orbit. 18
LEV	Lunar Entertainment Vehicle. 3, 4, 7–9, 11, 12, 15, 18, 19, 21, 22, 24–28, 34
LRV	Lunar Roving Vehicle. 10, 11, 16
LUM	Lower Unpressurized Module. 4, 9, 10, 16, 21–24, 27–31, 34
MLI	Multilayer Insulation. 22
NiTiNOL	Nickel Titanium Naval Ordnance Laboratory. 3, 15, 29
NRHO	Near-Rectilinear Halo Orbit. 26
OBC	OnBoard Computer. 12, 28, 30, 31
OWA	Ordered Weighted Average. 16
PDLC	Polymer-Dispersed Liquid Crystal. 3, 19, 20, 32
PSR	Permanently Shadowed Region. 7, 25
SEU	Single Event Upset. 31
SEV	Surface Exploration Vehicle. 10
SLM	Standby Lower Module. 23, 24, 27
SMA	Shape Memory Alloy. 15
TLI	Trans-Lunar Injection. 26
TMR	Triple Modular Redundancy. 31
UPM	Upper Pressurized Module. 2, 4, 9, 10, 21–24, 27–31, 34
USM	Upper Substitution Mechanism. 2, 3, 22, 24, 27

1 Overview

1.1 Purpose and Intended Audience

This document shall justify, outline, and give details about a new form of transportation on the lunar surface and its impacts on extraterrestrial mobility. This document is primarily intended for several actors in the space industry:

- Companies with plans to enter the lunar economy in the near future
- Automotive companies expanding to the space industry
- Commercial space station companies
- Anyone interested in space tourism
- Space agencies interested in a crewed lunar rover

By the end of this document, the authors intend to change expectations on how a crewed pressurized rover could come to fruition and why its modularity and focus on design (not only functionality) could spark a mobility and space tourism revolution.

1.2 Document Overview

The authors will first discuss the current lunar economy and space tourism and the system currently in place. The limitations of the current system and why a change should be made are then discussed. Next, they will discuss the concept for a novel system, including a background description, operational policies, a detailed description, and assumptions and constraints. The document will end with operational scenarios of the novel system, a summary of the novel system's impacts, and finally, some conclusions.

1.3 Stakeholders

Although the lunar economy and space tourism might not encompass many stakeholders as of today, it has a lot of potential towards the future decades (more discussed in section 2.1.1). Firstly, launch providers will be heavily implicated as they are needed to transport humans to the lunar surface. They will need to improve their current technologies and make them safe enough for the general public to have a higher degree of trust in them. Additionally, governments will also be involved in the future lunar economy as they try to provide their country with a significant presence on our celestial neighbor's surface. We also have to mention the scientific institutions, who will think of the possibility of stepping on the lunar surface as a gold mine. Stakeholders will also include all the new businesses associated with the new lunar economy (see Table 1). Least directly, but still notable, are everyday people across the world. Not only will they get a chance to explore the cosmos, but also reap the fruits of all the technological advancements made in order to colonize the Moon, some of which will be used daily on the Earth in just a few decades.

2 Current System

2.1 Background

2.1.1 Space tourism and future lunar economy

We find ourselves in another golden era for space: more Artemis missions are around the corner, SpaceX's Starship keeps improving test after test and more private companies are deciding to enter the space industry. Long gone are the days of the Cold War and the race to get first to the Moon. We are coming back in a few years with Artemis III, but this time we are going back to stay. As a result, this opens up a lot of opportunities and space tourism is certainly an important one. This sector already exists nowadays, with multiple private companies sending privileged individuals to space. As a matter of fact, its market size was valued at USD 848.28 million in 2023 [1]. Not only is its market size considerable for the limited reach the sector has, but it's also expected to grow to USD 27,861.99 million by 2032 [1].

As can be seen in Table 1, there are different actors in the space tourism sector, and most of them are in the US. Blue Origin and Virgin Galactic have already performed several suborbital flights for civilians while all the commercial space stations and space balloons are still in development. Most noticeably, Space Perspective is expected to launch their first passengers during Spring 2025, opening up a new way to explore the cosmos without having to feel the enormous thrust of a rocket while it tries to leave Earth's atmosphere.

Table 1: Private space companies and their crewed vehicles

Commercial Space Stations	Suborbital Flights
VAST's Haven-1 Space Station (VAST)	New Shepard (Blue Origin)
Axiom Space Station (Axiom Space)	VSS Unity (Virgin Galactic)
Orbital Reef (Blue Origin, Sierra Space)	Starship (SpaceX)
Starlab (Voyager Space, Lockheed Martin)	
Space balloons	Lunar surface
Spaceship Neptune (Space Perspective)	Toyota (Lunar Cruiser)
HALO Space	

Right now it can be hard to imagine a substantial lunar economy existing in the near future. However, it is by no means far-fetched to assume that the Artemis landings and the development of SpaceX's Starship will give it a considerable push. Participants in the new lunar economy could range from governments to scientific institutions to private companies. Find all the details of the future lunar economy in Table 2.

Table 2: Lunar goods and services organized by customer type (by 2040). Source: [2]

Governments and Scientists	Individuals and Businesses
<i>Human space exploration</i>	<i>Individuals</i>
Prolonged human presence on the Moon	Moon rocks
Research about human's health in space	Lunar tourism
Testing space technologies in situ	Artifacts made on the Moon to sell on Earth
	Lunar memorials (ashes on the Moon)
<i>Space science</i>	<i>Businesses</i>
Building telescopes without atmospheric interference	Helium-3 extraction for nuclear fusion
Lunar science	Extraction of precious minerals
Answering cosmological questions	Lunar manufacturing
	Waste disposal
	Lunar advertising
	Supercomputing and data storage
<i>Geopolitical Strength and Defense</i>	
Testing military technologies	
Lunar surveillance	
Permanent strategic outpost	

2.1.2 Lunar environment

To better understand how to design the LEV it is necessary to know the harsh conditions of the lunar environment, specifically at the South pole where the future lunar base has been assumed to be at (see 4.1.1). The most basic aspect to consider is that gravity on the surface of the Moon is roughly $g/6 = 1.6 \text{ ms}^{-2}$, meaning that objects there weight six times less than on the surface on the Earth. Also, there are large temperature changes as temperature can vary from -233°C to 123°C [3]. Specifically, the temperature differences of the lunar surfaces measured by Apollo between day and night are as high as 300°C . Additionally, the temperatures in PSRs are some of the lowest in the Solar System as they haven't seen sunlight in billions of years, reaching -250°C .

Over the last 50 years that we have had lunar samples and remote sensing data from the Moon, it is fair to say that there are two important processes that should be taken into account while understanding space weather on the surface of the Moon. The first one is the incoming flux of solar highly-energetic charged particles and their radiation which can definitely harm a human being if not enough protection is applied. Although cosmic rays from other stars and galactic bodies is also present on the surface of the Moon, there is not enough statistical evidence to support that they play a huge role with the available data [4]. The second process is the flux of hypervelocity impacts from micrometeoroids. As the Moon doesn't have an atmosphere, small meteoroids don't burn up or slow down via air friction, and can reach the surface of the Moon causing

damage. Additionally, they impact the surface and can produce ejecta, that could damage vehicles on the surface [5].

It is well known that the lunar surface is made up of regolith, whose small particles are known as lunar dust. Lunar dust particles have a mean diameter of 19 microns and their main components are silicon dioxide or SiO_2 (44.72%) and aluminum oxide or Al_2O_3 (14.86%)[6]. They have several known properties, like being magnetic and highly porous. These characteristics need to be taken into account as they may impact decisions on the materials and the technologies used for the LEV. At the same time, some experiments suggest that there might be sunrise-driven movement of this dust. Fully understanding this phenomenon might be crucial in the future of mankind's expeditions to our planetary neighbour as lunar dust has such small particles that they can get inside of almost anything.

Another point to take into account when studying the lunar environment is its seismic activity. All the data we have regarding this topic comes from the Apollo missions' seismometers. Here are the different types of moonquakes:

- **Thermal moonquakes:** They take place thanks to the continuous heating and expansion of the crust of the Moon. However, they have the lowest magnitude of all moonquakes and are generally not considered dangerous for any humans stationed on the lunar surface.
- **Deep moonquakes:** These still have a small magnitude (≤ 3)¹ and are associated with tidal forces (due to Earth's gravitational effect on the Moon).
- **Meteoroid impacts:** These consider all types of meteoroids, not only the small ones. There were more than 1700 events during 1969-1977 with masses from 0.1 kg to 100 kg, leading us to think that coming up with solutions for these impacts is important, just as we mentioned in the second paragraph.
- **Shallow moonquakes:** These are the most serious ones as they can reach magnitudes of >5 . with focal depths as high as 50 km and as low as 200 km. Although it's uncertain, the cause for these moonquakes could be associated to boundaries between different surface features. These are considered the riskiest for any human activity on the lunar surface.

2.2 Description of Current System and Change Justification

In the status quo, there are not many proposals for pressurized lunar vehicles and none have actually been on the lunar surface. This is due to the lack of lunar infrastructure and the immense cost of launching mass into space. However, JAXA and Toyota are developing a crewed pressurized rover named Lunar Cruiser that will be used by astronauts in the Artemis program in the near future. It will allow 2 astronauts to take drives of between 30 and 45 days on the Moon as they fulfill their scientific duties. It is expected to be on the Moon sometime in the 2030s.

For the Lunar Cruiser and all the manned spacecraft that have been used to this date, design and aesthetics have not been taken into consideration. Rightfully so, engineers have primarily cared about the functionality of the vehicles and their associated safety. However, as we enter this new golden era of space, passengers inside these vehicles will start looking for a more comfortable and premium experience. The authors have identified this as an area of improvement for all space tourism ventures, including lunar rides. A tour on the lunar surface should make passengers feel young again, limitless and fearless. The best way to convey emotions and feelings is through art and design.

IBO Engineering has been offering services in the automotive industry for over 25 years, treating the 2 most important areas in the sector: design and engineering. It's clear that the aerospace industry has one of the best technical and engineering supports one could ever hope for, but design has not been as important as it will be in the future. There have been a few private space companies that have started to give it the importance it needs. For example, Axiom Space has been working with visionary creator Philippe Starck to develop their commercial space station. Philippe is a French designer who is known for his work in interior and product design and architecture. Another example is Voyager Space, as Hilton has recently been named their official hotel partner as they develop the commercial space station Starlab. Again, everything points to improving the passenger's experience in space. Although there are a few actors in the industry already thinking about this problem in particular, there is still a lot of work to be done, and none of them are working on lunar products. As a result, the authors feel strongly about this change and would like to propose a system that satisfies both the technical and design needs.

¹Expressed on the Richter scale.

3 Concepts for the proposed system

3.1 General description

The proposed system is a **lunar pressurized off-road** vehicle named Lunar Entertainment Vehicle (**LEV**) and its main purpose is to bring space tourists on a ride on the lunar surface. It will be an extremely safe, comfortable vehicle divided in 2 parts: the Lower Unpressurized Module (**LUM**) and the Upper Pressurized Module (**UPM**). **Modularity** is probably the most important feature of the LEV, as it will enable different upper modules, each made for fulfilling different purposes (construction, science, entertainment...) to be joined to the LUM, which will lead to less costs and better use of the available material.

Table 3: Division of subsystems among the UPM and LUM.

UPM	LUM
Advanced navigation computer and sensors	Onboard Computer unit and navigation sensors
Emergency battery unit	Power electronics
Life Support	Power source (hydrogen-oxygen fuel cells)
Upper module thermal control	Lower thermal control
Waste management	Rudimentary communications
High bandwidth communications	External lighting
Lights (interior and exterior)	Suspension
	Motor/wheel assembly
	Steering system and actuators
	Auxiliary subsystems

The LEV will have a maximum capacity of 7 passengers, with 1 of them being a multi-functional profile responsible for driving, health emergencies and others. As we expect the LEV to be driven **autonomously**, the multi-functional profile will only have to drive in specific scenarios where manual drive is needed (which will be very rare and uncommon). Every ride will bring **6 space tourists**. The duration of each trip is not supposed to last more than a day as the maximum driving time will be **10 hours**. There will be times where the vehicle will not be moving for the passengers to have time to look at certain lunar features. The space tourists will have to wear **pressure suits** during the complete duration of the mission such that they can be safe in the case of an unexpected rapid cabin depressurization event. The suits will be comfortable to wear, with the ability to be used without their visor being closed during nominal missions. Additionally, they will be aesthetically pleasing, contributing to create a more immersive experience. The feasibility of creating such a pressure suit has been proven in the recent years by SpaceX [7]. In the case of an emergency the LEV can maintain 8 people alive during a maximum of 48 hours (see 4.3). **No EVAs** will be performed by the space tourists. This is mainly due to the fact that they are not trained astronauts, and performing a lunar EVA is a highly specialized task that requires extensive training, which would raise the costs enormously. The possibility of integrating EVA capabilities into future rovers, however, will be considered for future mission concepts of a different nature.

3.1.1 Vehicle mass

The initial mass of the vehicle was estimated as the starting point for designing the rest of the vehicle, as it impacts several design parameters. In order to do so, the system's requirements were taken into account: the target crew and operational range that were set as initial requirements, were found to have a strong correlation with the mass of similar manned pressurized rover proposals for space exploration. It was then deemed that a statistical model of the vehicle's mass according to these key design parameters could return a reference design mass as a starting point.

Data from similar rovers was gathered, mainly from sources [8], [9] and [3], and the rovers that deviated from the LEV's core concept were discarded as outliers in order to create a rover dataset that is representative of the LEV's essence.

Table 4: Dataset of mass and related design parameters of similar rovers.

Name	Crew	Range (km)	Mass (kg)
Rover First	2	80	4300
USRA Creel Studies	4	500	6200
USRA Bhardwaj Studies	4	2000	7000
Arno	3	100	6080
Morphlab	4	1000	3700
NASA CR 192033	4	1000	7015
NASA CR 192034	2	3192	6197

The data presented in table 4 can be used to predict the mass of a similar rover given an input crew and operational range. Performing for a crew of 6 and a total range of 350km (or an operational radius of 175 km), the mass of the rover can be expected to be of 6708.4 kg. It should be taken into account that many of the rovers being used as reference points come from older proposals. Therefore, engineering advancements should accommodate a reduction in the estimated mass. For this reason, the initial estimation for the **mass of the rover** will be of 6000.0 kg or **6 metric tonnes**. The weight distribution between the upper and lower modules will be taken, for the preliminary design phase, by taking the weight distribution of the NASA SEV vehicle, which means that the UPM is expected to weigh approximately 2000.0kg or **2 metric tonnes**, and the LUM is expected to weigh 4000.0kg or **4 metric tonnes**.

3.1.2 Vehicle dimensions

As a way of guiding the design process, a range of possible dimensions within which the final rover will have to fall into has been given. Some of the dimensions are set by operational constraints, such as the ground clearance or the wheel track, while other dimensions have either been estimated from similar vehicle proposals, or determined via a preliminary packaging process using CAD software, using possible interior and subsystem layouts. Additionally, the modularity of the vehicle concept also plays a role when defining the overall sizes.

The dimensions of the UPM shall fall within the following ranges ².

- Height: 2.5 - 3.1 m
- Length: 8.0 - 9.5 m

The UPM's width should not be larger than the wheel track, while a minimum of 3.5 m of width is recommended from the point of view of the passengers' comfort.

In order to house some of the upper module's subsystems, an internal volume of 2.41 m^3 must be included. This volume houses the technical machinery of the UPM and can be split into different sub-volumes as low as 0.3 m^3 . It can also accommodate various shapes to fit the designers' aesthetic and interior layout needs.

The LUM's dimensions are given under the framework of a large rectangular chassis where most of its subsystems are housed, together with the motor-wheel assembly being mechanically connected to said platform via a suspension mechanism.

The chassis portion of the LUM shall be constrained to the following measurements, which will ensure an adequate integration with the upper module, as well as enough space for housing its subsystems:

- Minimum height (thickness): 0.43 m
- Maximum length: body length
- Minimum top section area (width $\times$ length): 10 m^2
- Ground clearance: 0.75 - 1.0 m

The track of the vehicle has been determined by analysing its stability under lateral accelerations, and ensuring a safety factor and steering capabilities on par with the Lunar Roving Vehicle (LRV) first deployed in the Apollo 15 mission. This safety factor can be defined with the ratio between the fictitious displacement of the vehicle's center of gravity that causes a roll torque equivalent to the contributions of the weight on

²The minimum values of the intervals are hard constraints, whereas the maximum ones are soft constraints which could be surpassed if properly justified.

the real center of gravity and the centripetal acceleration under turns, to the half of the vehicle's track δ_{cg} . When this ratio is larger than 1, the vehicle will tip over due to excess lateral acceleration. A similar ratio shall be used for the LEV.

The maximum lateral acceleration of the LRV can be computed using the turn radius and velocities given in figure 3-10 from reference [10], which corresponds to 2.04 m/s^2 . Taking the LRV's track into account, it can be seen that its CG's fictitious displacement is at the 82% of the roll-over limit, as has been computed using equation 1³.

$$\delta_{cg} = \frac{h_c}{g} a_{lat} = \frac{h_c v^2}{g r_t} \quad (1)$$

The height of the LEV's CG has been preliminary estimated by applying the center of gravity formula for a particle system considering the upper and lower modules' weights, where their respective CGs have been placed at the vertical position corresponding to half of their heights. This returns a value of $h_c = 1.38 \text{ m}$. If the maximum lateral acceleration and the turning or roll-over safety margin are selected to be the same as those from the LRV, **the minimum track of the LEV results in 4.3 meters.**

3.1.3 Energy requirements and storage

Several types of energy sources have been considered for powering the rover. After contrasting each option's pros and cons against the scenario detailed in 3.2, as well as analyzing their mass efficiencies, it has been determined that a power system based on primary Hydrogen-Oxygen fuel cells as the main power source together with a Li-BCX emergency battery is the best solution for powering the rover.

The first step in this consideration was studying the energetic and power requirements of the rover. The total power required to operate the rover can be broken down into a static and driving power, the first of which can be considered mostly constant through the rover's operation and is used for powering the different subsystems. The driving term comprises only the energy that is used for moving the vehicle across the lunar soil.

In the case of a wheeled vehicle, the total power necessary for driving can be obtained by using a friction coefficient f_r which relates the vehicle's weight with the force needed to overcome its friction. According to source [9], this friction coefficient can be approximated by a value of $f_r \approx 0.2$ for lunar driving. The friction force F_f needed for driving the vehicle at a constant speed on a flat surface can be obtained as follows.

$$F_f = f_r M g \quad (2)$$

This is independent of the number of wheels, as the total reaction force exerted by the wheels must always equal the vehicle's weight (when driving in a level surface), and therefore the sum of the friction forces of all wheels is the same as the value given by equation 2. The power used for driving comes as the product between the force used for driving and the speed at which the vehicle moves, as given in equation 3. In this manner, the peak driving power at the wheels can be estimated.

$$P_{drive} = F_f v \quad (3)$$

In order to have a more complete power model, the static power loads must also be considered. These include lighting, life support, guidance and navigation, steering, thermal control, etc, and their values will be taken from the estimations made in reference [3] after extrapolating the ECLSS power for use with 6 people and doubling the steering power, as suggested by [11]. These values are collected in Table 5.

From these values, two kind of power consumptions can be characterized:

- **Driving static power** (P_d) : all of the above. Must be accounted for while driving
- **Stand-by power** (P_s) : only thermal control, ECLSS, and external and internal power. Must be accounted for when not driving.

Additionally, an efficiency factor η must be incorporated to represent the efficiency losses in the mechanical transmissions as well as the power distribution. According to reference [12], about 65% of the energy stored

³Where a_{lat} is the lateral or centripetal acceleration, h_c is the height of the real CG, v is the velocity, r_t is the turn radius and g is the gravitational acceleration on the lunar surface

Table 5: Power consumption by subsystem, adapted from [3]

Subsystem	Power (W)
Thermal control	800
Operations	600
ECLSS	300
External power (lights, sensors ...)	400
Internal power (lights, GNC, OBC ...)	200
Steering power	500

in an electric vehicle's battery is used, with the rest being lost to electrical and mechanical inefficiencies. The peak power, which will occur while driving at maximum speeds can be found as:

$$P_{peak} = \frac{1}{\eta} (Mgf_r V_{max} + P_d) \quad (4)$$

The energy that must be consumed for a specific mission profile can be found by integrating the consumed power over time. In order to do so, the vehicle's velocity will either be a constant value, corresponding to the operational speed, or zero, in which case the standby power is consumed. Taking this into account, two design times can be defined: the mission driving t_d , and the stand-by time t_s . This results in equation 5.

$$E = \frac{1}{\eta} [(Mgf_r V_{nominal} + P_d) t_d + P_s t_s] \quad (5)$$

A safety factor is used for evaluating the peak power and the total energy needs. Drawing inspiration from aerospace structure designs, an ultimate factor of safety is used for determining the worst-case power loads as well as for the energy requirements. It should be taken into account that the power loads are generally well understood, and power limits can be set to the power-train to avoid excessive loads to the power sources. The uncertainty that shall be accounted for with a safety factor is that of the terrain. With the aim of characterising this, a safety factor can be computed such that it increases the power draw for climbing up a hill, with the safety factor being a function of the steepness of said hill. In this manner, the total energy requirements that will be assessed will assume that the vehicle will be constantly climbing a 3^o steep hill. This leads to an energy safety factor of 1.26 according to equation 6 where α is the hill's angle.

$$SF = 1 + \frac{1}{\eta} \tan \alpha \quad (6)$$

After taking all of these factors into account, the following operational capabilities and energetic requirements are given in tables 6⁴ and 7 respectively.

Table 6: Operational parameters of the LEV rover on the Moon.

Operational capability	Value
Max speed	30 km/h
Nominal speed	20 km/h
Driving time	10 h
Max Mission time	48 h
Max Operational Range	224.63 km
Nominal Operational Range	219.85 km

Table 7: Energy and power requirements of the LEV rover on the Moon.

Energetic need	Value
Peak Power	26.93 kW
Nominal Power	19.21 kW
Stand-by Power	1.80 kW
Total Energy	362.32 kWh

⁴Operational ranges assume driving on flat terrains under uniform speeds.

Given these energetic requirements, the energy source can be sized. First, the mass and volume of the fuel cells has to be found, for which the power (in kW) and energy (in kWh) mass and volume densities (in kg and l) are given by equations 7, 8, 9 and 10 respectively, which are derived in reference [13].

$$\rho_p^p = \frac{3661 \frac{1}{t_{dis}} \eta_e \mu}{3661 \eta_e \mu (V_d I_d A_m t_{dis})^{-1} + 83.3 RT + \mu} \quad (7)$$

$$\rho_e^p = \frac{3661 \eta_e \mu}{3661 \eta_e \mu (V_d I_d A_m t_{dis})^{-1} + 83.3 RT + \mu} \quad (8)$$

$$\gamma_p^p = \frac{1}{t_{dis} RT (43.93 \eta_e P_t)^{-1} + (V_d I_d A_v)^{-1}} \quad (9)$$

$$\gamma_e^p = \frac{1}{RT (43.93 \eta_e P_t)^{-1} + (V_d I_d A_v t_{dis})^{-1}} \quad (10)$$

Most of the design parameters for the fuel cells have been chosen in accordance with the design reference [13], and are listed in table 8. The last variable left is the design discharge time t_{dis} .

Table 8: Fuel cell design parameters.

Fuel Cell Coefficient	Value
Energy conversion efficiency η_e	50%
Tank figure of merit μ	1800 atm/liter · kg
Tank pressure P_t	400 atm
Discharge voltage V_d	0.75 V
Discharge current I_d	1 A/cm ²
Contact area per mass A_m	1540 cm ² /kg
Contact area per volume A_v	215 cm ² /liter
Gas temperature T	200 K

In order to correctly size the fuel cell, three discharge times will be tested: a nominal or driving discharge time consisting of the 10h of driving, a total discharge time of 48h of the entire mission's duration, and a peak discharge time of only 20 minutes. For each design discharge time, three operational scenarios will be tested, and the one which needs the maximum mass will be stored, as this would correspond to the minimum mass needed for meeting all the power supply demands under the given design parameters. Then, the design discharge time whose maximum mass is the smallest will be stored as the optimal fuel cell needed to meet the most demanding load scenario. Applying this procedure yields the fuel cell parameters specified in table 9.

Table 9: Fuel cell specifications

Fuel Cell Parameter	Value
Weight	384.21 kg
Volume	912.12 liters
Peak power	36.68 kW
Total energy	366.79 kWh

An emergency energy storage is also necessary in case of any errors in the fuel cells or in the electrical connections between the modules. Li-BCX non-rechargeable battery cells can be used for this application since their usage will be exceptional. In the case of chemical batteries, their mass and volume densities depend on their chemistry, which is specified in table 10, as well as on their final packaging, for which a packaging efficiency of 67.7% will be taken as reference (matching that of the *Mars Exploration Rover*).

Table 10: Li-BCx Cell Parameters, obtained from [14].

Li-BCX Cell parameter	Value
Mass power density	150.0 W/kg
Volume power density	337.0 W/l
Mass energy density	414.0 Wh/kg
Volume energy density	930.0 Wh/l

The design loads for these batteries include being able to supply the stand-by power, as well as the stand-by energy across 38h of stand-by time. The most restrictive case is stored, and the corresponding emergency battery pack is obtained, as specified in table 11.

Table 11: Emergency Battery Pack specifications

Battery Pack Parameter	Value
Weight	189.90 kg
Volume	73.45 liters
Peak power	19.28 kW
Total energy	53.20 kWh

3.1.4 Wheels

Circular-shaped wheels

The wheels of the rover are one of its most important components as they are the source of contact with the ground. Their correct operation includes achieving good traction while keeping a low rolling resistance, being able to ride over obstacles without causing wheel failure, being durable to continued use while under extreme conditions, and transferring the vehicle's loads to the ground effectively, among many others.

Many different wheel concepts have been proposed for planetary exploration. These are inherently different from typical Earth wheels due to the different conditions that are present in the lunar surface, as highlighted by [15], which include:

- Wide variations in temperature (-230 to 130 °C).
- Exposure to direct solar radiation.
- Operation in a hard vacuum.
- High vehicle loads despite being subjected to a lower gravitational field.
- Contact with loose terrain composed of regolith and dust with varying soil properties.

Wire mesh wheels have been shown to be the optimal option for manned rovers, since they have to bear higher loads than unmanned ones while providing good handling characteristics and a smooth ride. A trade study comparing various lunar wheel concept proposals [15] shows that this option dominates in terms of keeping a low weight, providing comfortable ride experiences and overcoming obstacles. These wheels are built using metallic wires that are crimped and weaved into a cylindrical mesh, that then gets shaped into a toroid that gets secured into the rim.

Obstacles that have acute edges may damage the wheels, by warping the metal wires locally into the plastic deformation region, yielding an unusable wheel. While this has been overcome in the past by adding sheet reinforcements to stiffen the wheel as well as bump stops to avoid excessive strains on the mesh, recent advances in material science has allowed for shape memory alloys to be used instead of the old steel alloys. When properly conditioned, these alloys can enter a state of superelasticity, where they are able to sustain elastic deformation way past their usual elastic limit. This avoids the need of adding reinforcements or stops to the rims, and largely improves the mechanical reliability of wire mesh wheels.



Figure 1: Sample of a NiTiNOL wheel assembly by Smart Tire. Source: [16]

The specific SMA or Shape Memory Alloy that has been selected for the LEV's wheels is NiTiNOL, which also exhibits super-elasticity. A NiTi alloy can be engineered to operate in an austenitic state, that is, at a temperature above that in which martensite starts forming in a cooling process (commonly known as martensite start temperature). In its austenitic form, only one possible crystallographic configuration exists - a simple lattice structure which contains the material's original shape. If stresses are applied to such a material, stress-induced Martensite forms (as if the metal were being cooled down), making it easily deform into an arbitrary shape, since Martensite deforms by twinning, where 24 different crystallographic configurations can accommodate deformations before slip or dislocation mechanisms deform it permanently. When stress is released, the martensite transforms back into austenite, making the material recover its original shape as it regains its single possible lattice configuration, giving rise to the super-elastic effect. Wire mesh rover wheels made out of NITINOL have already been developed and ground-tested on Earth, reaching loads of more than 4.4 kN, making this technology more than feasible for the LEV vehicle.

Preliminary analysis using a spring-based contact patch model of the wheel and the Bekker model for soil sinkage has shown that the **minimum wheel diameter** should be **of 0.85 m**, with a **recommended wheel diameter of 1.65 m**, where both figures would achieve a loose soil sinkage of roughly 3 cm, as indicated by reference [3], where the minimum and recommended values correspond to the cases of the wheel undergoing poor and nominal performance respectively.

Triangular-shaped wheel tracks

The use of tracks for off-road locomotion is not new on Earth. Tracks, when compared to wheels offer a wide array of advantages for traversing difficult terrains. Most of these advantages originate in the fact that tracks have much larger contact areas with the soil than traditional wheels, which lowers the slip coefficients and makes the energy transfer to the ground more efficient, giving the vehicle the ability to climb steeper slopes. In addition, the larger contact area makes the rover more capable at driving over obstacles.

However, using tracks instead of wheels, usually offers some disadvantages. The biggest one is the increase in power draw for driving since, even though the slip is reduced, the friction with the ground is also increased. Additionally, more complex mechanical systems are required to maintain the track's shape and transfer power to it, meaning that the overall mechanical efficiency is likely to drop. The power required for turning can increase significantly if differential steering is used which, as was already described during the wheel selection process, can increase the peak power required by a factor of 2 when moving at very low velocities.

Despite this, some researchers have been able to design small and lightweight tracks using metal meshes that have achieved lower slip ratios while also reducing the power draw when compared to wheels of similar characteristics. These have then been arranged into four sub-track units to form a tracked rover with an otherwise conventional configuration, which seems to be promising for enhancing mobility and efficiency on the lunar terrain in mild slopes (see source [17]). Additionally, this configuration includes conventional steering of the crawler modules, making the vehicle even more power efficient.

One of the biggest limitations when designing wheels for extraterrestrial mobility applications is the material selection for the tires, especially in atmosphere-less and high-radiation environment conditions such as the lunar surface, where the temperatures range from -133 °C to 121 °C. After an internal study, all studied rubber materials fail to retain adequate mechanical properties across the extreme temperature variations of lunar conditions. Their glass transition temperatures are too high, and their rigidity degrades at insufficiently high temperatures. While some rubbers exhibit promising UV resistance, none meet the demands of this application. Although advancements in polymer science may eventually yield a suitable rubber compound,

metals remain the only viable option for now. In particular, NiTiNOL meshed treads present a promising alternative, offering durability and adaptability in the harsh lunar environment.

Such treads have already been studied for applications in lunar mobility. One such approach has been studied in source [17] which creates track modules or crawlers made out of off-the-shelf metal mesh (which can be presumed to consist of steel alloys) together with L-shaped lugs made out of brass. This configuration has shown to deliver lower slip ratios than a similar rigid wheel at lower powers, and has been used in a configuration similar to IBO's LUM rover, using more than two crawler modules in one single vehicle.

The use of metallic treads can enable a much more versatile material selection, since most metals resist the operational temperature ranges required while retaining their mechanical properties, and can withstand high doses of UV radiation successfully. While the tracks may mechanically degrade over time due to their lower breaking deformation when compared to rubber materials, proper maintenance can be performed to avoid this from becoming an issue. Even then, mechanical degradation while riding over lunar regolith is expected to be low. Special alloys such as NiTiNOL could be used to further avoid problems with mechanical degradation, especially when driving over rough or hard terrains, as their super-elasticity could bring the metal treads a much greater breaking deformation, making sharp features on the ground much less threatening to the structural integrity of the treads.

The study of both the multi-tracked vehicle concept and material selection confirms the viability of the proposed rover, provided the tread material is changed to a metallic alternative. The multi-tracked system proves to be an effective and relatively efficient locomotion strategy for traversing lunar terrain, being comparable in terms of energy draw in low slopes. The use of multiple small tracks significantly enhances mechanical grip. Additionally, integrating conventional steering further improves efficiency, especially at low speeds. The mesh treads seem to provide a simple yet effective approach to manufacturing the tracks.

Suspension for the wheels ⁵

The inclusion of a suspension system must allow for greater ride comfort, improved handling on smooth terrain, as well as increase the rover's mobility on complex terrain geometries. Such a system will inevitably add mass and volume to the vehicle. Many suspension systems have been developed over the years for Earth-based vehicles, as well as for extra-terrestrial exploration vehicles, with one of them (the LRV) being manned. It should be noted, however, that the LRV has been one of the only rovers to include a more traditional convention system, mainly due to the high speeds and ride comfort aspects playing a major role.

Taking these factors into account, various suspension mechanisms for off-road vehicles have been compared using the OWA method, taking different considerations into account such as weight, maintainability, ride comfort or terrain adaptability into account, among others. The most suitable suspension mechanism has turned out to be an independent **Double-Wishbone suspension**. Further analysis using suspension design software has lead to the suspension geometry outlined in figure 2, which allows for large contact patches despite wide suspension deflections and body rolls. It should be mentioned that a study of state-of-the-art IWS systems where the suspension is embedded inside the rim of the wheel were also studied, but came in inferior place most of the studied traditional suspensions.

⁵For the wheel tracks, the suspension mechanism would be placed inside the triangular-shaped wheel tracks. The specific type of suspension will be assessed in future reports.

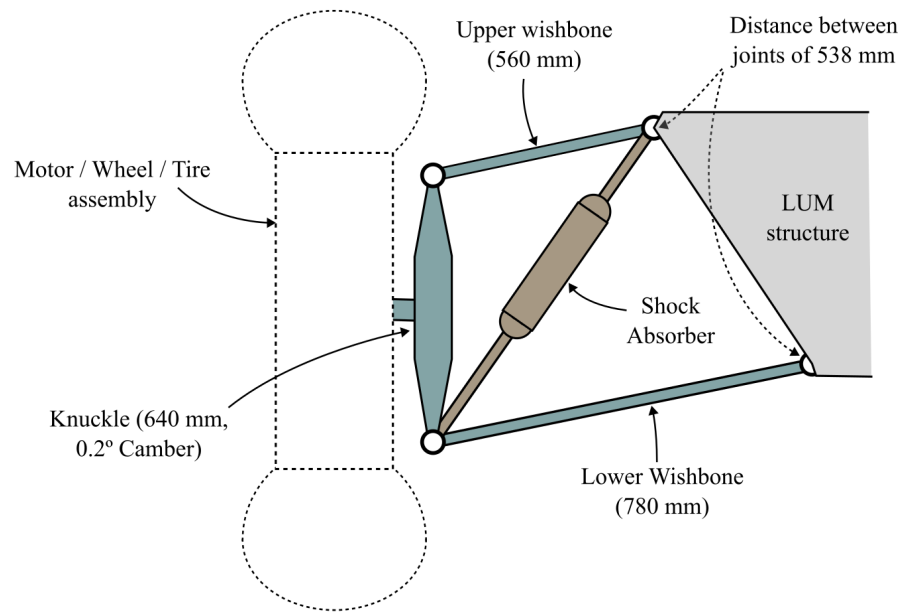


Figure 2: Front view of the Double-Wishbone suspension geometry with its elements and their lengths specified.

The stiffness of the springs was also preliminarily studied using a one-sixth suspension model of the vehicle, which allows for the tuning of the basic suspension parameters such as spring stiffness and damping coefficients while taking into account the sprung vs unsprung mass, as well as the tire's stiffness.

3.1.5 Access point

Only one access point at the rear end will be implemented and will be used as an entry point (from the lunar habitat) and an emergency door. In the case of emergency, two vehicles will dock to each other (see Section 4.3). Note that the IDSS has a diameter of 1.255 m and a transfer passageway of 0.8 m. Although one might think that this opening is too small for tourists to go through, we decided to implement this mechanism for our access point because of the following 2 reasons:

1. **Security and standardization:** All international space vehicles have always tried to use the same docking system as it enables international cooperation and could be useful in cases of emergency. All the countries using the same docking system makes a lot of sense, which is why we are assuming that there's a high chance of the IDSS also being used on the surface of the Moon. Additionally, this system has been used for decades now, making it secure and reliable.
2. **Immersive experience:** Although the clients are expecting a luxury experience because of the highly expensive tickets, going through an actual IDSS might be an interesting activity as it would make the experience much more immersive. After all, they will be on the surface of the Moon.

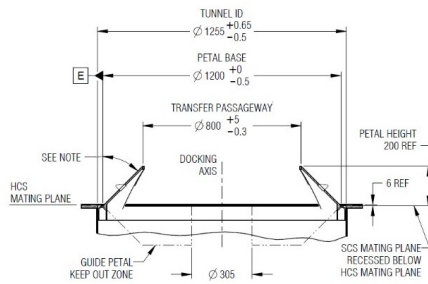


Figure 3: Dimensions of the IDSS. Source:

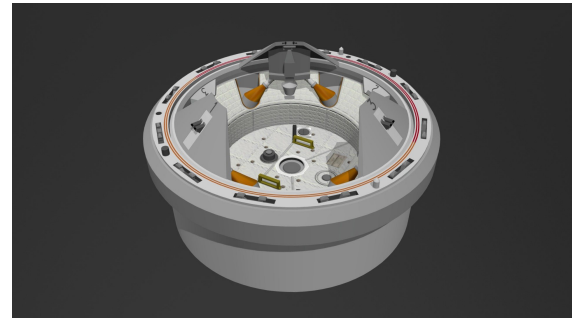


Figure 4: Digital construction of the IDSS

Figure 5: Access point proposal

3.1.6 Windows

When designing the windows of a lunar pressurized vehicle several factors need to be taken into account. Some of these include pressure difference, radiation, illumination, small meteorites and debris among others.

Two flight-proven large window design for low to zero-pressure environments have been taken as references for the window design. These include the windows from Space Perspective's stratospheric balloons⁶, due to their prominent use of large windows in their capsule, and the windows on the ISS, as it presents several large windows that have been operating for decades in LEO. Space Perspective's windows are particularly interesting because they will be the largest ever flown with humans onboard to near-space when they start with their commercial flights in 2025. Their height is 132 cm. Although the information about their width is not public, we will take it to be 50 cm as this number is a bit higher than the average male body width.

One might wonder why this comparison makes sense, as we all know that these balloons are not exactly reaching space. In Space Perspective's case, they announced that their flights will reach 30 km. If we derive the formula for atmospheric pressure and its dependence on height, we find that it decreases exponentially. Specifically, knowing that the pressure at the sea level is 1 atm, then the temperature at 30 km is -45 °C, which means that $P_{30} = 0.011$ atm, which makes the relative pressure and, therefore, the loads on the window, effectively equivalent to those experienced on the Moon. While Space Perspective's windows have

Table 12: Space Perspective's windows specifications

Dimensions (cm)	
Height	132.0
Width	50.0

inspired the necessary confidence for the large sizing of LEV's windows, their core design and architecture

⁶<https://spaceperspective.com>

has been largely inspired by that of the ISS. The final cross section of the LEV's windows is composed of the following layers⁷:

1. Mint tint

A very thin tinting film composed of a combination of ceramic and carbon nanoparticle layers film is proposed for protecting the LEV's interior from sunlight. This includes UV protection for the passengers thanks to the ceramic particles, and infrared absorption capabilities of the carbon particles for thermal management. Additionally, both layers have shown to be highly durable, making them suitable for long-term use. These layers also have the advantage that they do not interact with electromagnetic fields (as the metallic alternatives do), making them highly suitable for correct vehicle operations [18]. As a starting point, a VLT (Visible Light Transmission) coefficient of 60% is selected, with the PDLC layer of the windows adaptively blocking the rest of sunlight for optimal visibility conditions.

2. Outer meteoroid/debris pane (0.93 cm) [19]

Protect pressure panes from impacts due to meteoroids and debris.

3. Redundant pressure pane (2.54 cm)

Just in case there's an issue with the primary pressure pane.

4. Primary pressure pane (2.54 cm)

Its main function is to deal with the difference of pressure between the pressurized cabin and outer space

5. PDLC (0.1 mm)

The LEV will experience different amounts of radiant flux from the Sun as it navigates on the surface of the Moon (more on this in 4.1.1). As a result, the space tourist might want (or even need) to change the transmittance of incident light on the window. This can be achieved with a PDLC, which is a homogeneous mixture of polymer and liquid crystal microdroplets [20]. These microdroplets are positive birefringent nematic liquid crystals smaller than a micron.

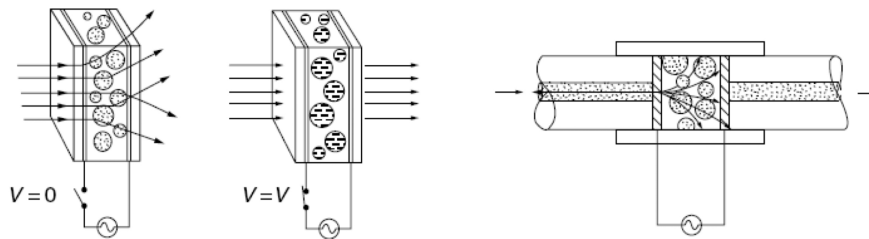


Figure 6: Different voltage settings for a PDLC. Source: [20]

As we see in Figure 6, these are the extreme configurations of a PDLC:

- $V = 0$

In this case, the external electric field is zero, therefore implying that the orientation of the molecular axes of the nematic liquid crystal in the microdroplet is arbitrary. Consequently, the microdroplets become scattering centers and the transmittance of the incident light is at its minimum.

- $V = V_{\max}$

When the transmittance of incident light through the PDLC is at its highest. In this case an external electric field is applied, which results in an alignment of the crystal axis because of the positive birefringence. Incident light perpendicular to the PDLC with any polarization state will see the ordinary refractive index n_0 .

Changing the amount of voltage applied, one can go from the first configuration to the second, which in turn changes the amount of light that passes through the PDLC. It has been suggested that the optimal thickness of this layer should be 0.1 mm, according to [21].

6. Inner scratch pane (1.14 cm) [19]

It protects the important pressure panes as astronauts inside the LEV could damage them with an unintended hit.

⁷it should be noted that this design is inspired by ISS Cupola, which at the moment had the biggest windows ever flown to space

In order to verify that the structural integrity of the windows when under internal pressure loading is guaranteed, simple structural analysis can be performed on a simplified window panel with the reference measures of that produced by Space Perspective, being a rectangle of 1.25 by 0.5 m. The maximum stress that occurs under such conditions when the edges are simply supported can be computed as follows [22], with a and b being the sides of the rectangle, t being the window's thickness, and β being a constant that depends on the a/b ratio, being equal to 0.68 in this window's case.

$$\sigma_{\max} = \beta \frac{P_{\text{atm}} \min(a, b)^2}{t^2} \quad (11)$$

By solving equation 11 for the panel's thickness, the minimum necessary thickness of the plate before reaching the tensile stress can be computed for any given pressure, or alternatively, the pressure that a window of a given thickness can be computed. The mechanical properties of the glass, which consists of fused silica, can be obtained as the average values found in reference [23]. The maximum pressure that the space perspective windows could handle given a thickness of 2.54 cm is of 3,6 atmospheres, making these pressure panes compliant with the rest of safety margins being used.

Taking all of these factors into account, the total thickness of the windows will be approximately **15 cm**.

Sunlight's impact on visibility

The lack of atmosphere on the moon's surface means that the Sun's radiation is not additionally attenuated through the scattering and absorption mechanisms of the Earth's atmosphere. Approximately 18% of Sunlight's energy is absorbed by Earth's atmosphere before reaching its surface [24], which conditions it to the radiation levels that we are accustomed to. This absence of light attenuation was noticed on the moon's surface where, despite the EVA suits' helmets already including visors and sunshades, Apollo astronauts noted *"lunar surface visibility was not too unlike Earth visibility, with the exception that the sun was extremely bright"* [25]. The surface brightness is also accentuated by the increased albedo of the lunar terrain.

An important consideration for future lunar missions is the fact that the lunar South Pole is subject to some of the harshest illumination conditions in the Solar System, as the 1.54° inclination of the Moon's rotational axis relative to the ecliptic pole ensures the South Polar surface region never exceeds a Sun angle of a few degrees, leading to shadow detail loss in Sun-down positions, and direct solar viewing in Sun-up orientations.



Figure 7: Photo from EVA 1 of Apollo 12, showing the glare in the up-Sun direction. Source: [26]

These visibility conditions will be prevalent in the operational scenario in which future rovers will operate, and a solution that enables improved visual comfort must be sought. The inclusion of the last translucent protective layer, together with the PDLC glass in the windows' architecture will serve the purpose of fading the incoming light to the passengers such that they can observe the landscape and the terrain's features with much greater visual relief. Additionally, as previously mentioned, these will provide protection from high-energy and near-visible electromagnetic radiation such as UV, and reduce the heat that enters the vehicle's interior.

3.1.7 Lateral and roof cross sections

The pressurized section of the UPM will consist, structurally, of a composite monocoque structure, with additional external layers that will provide protection from environmental dangers, as well as make the structure tolerant to the environmental threats posed by the extreme temperature changes, as well as offer radiation and thermal protection to the crew. Table 13 outlines the architecture of all the layers that make up the UPM's structure, which are then explained in greater detail.

Table 13: LEV chassis' shell composition. Source: [3]

Layer	Material	Thickness (cm)
1, 2 and 3	Graphite/Epoxy-Nomex Sandwich-Structured Composite	0.25, 1.50 and 0.25 respectively
2	Honeycomb Core	1.50
3	Graphite/Epoxy	0.25
4	Elastomer	0.10
5	MLI	3.00
6	Spacing	4.00
7	Aluminum	0.10
Total		9.2

Layer 1: Graphite/Epoxy-Nomex Sandwich-Structured Composite

The innermost layer is a structural one, which consists of a Sandwich-Structured composite panel with its outer layers made of 0.25 cm thick Graphite/Epoxy composite material and a low-density Nomex honeycomb inner core. The choice for this architecture as the structural layer of the pressurized module is mainly due to the enormous increase in rigidity achieved given such a low material weight. This is achieved by increasing the local moment of inertia of the sections of the structure through separating the composite layers from one another, while having the inner core material transfer loads between them without adding excess weight. Such a structure is much better suited for facing bending loads than an equivalent structure with a single composite sheet of a thickness the sum of both external sandwich layers.

As temperature in the lunar South pole can vary a lot in a short period of time, the material on the outside of the LEV will expand and contract due to these thermal variations. As a result, special interest is given in materials whose coefficients of thermal expansion are low ⁸. Under this lens, a Graphite/Epoxy composite layer presents itself as a suitable material, as it has an almost negligible coefficient of thermal expansion. It is made from graphite fibers embedded in an epoxy resin matrix, making it strong, lightweight, stiff and thermally stable. The inner core material should follow suit, while focusing more on being lightweight. This can be accomplished with a honeycomb panel made of NOMEX, a material that has a high thermal tolerance and is strong and lightweight.

In order to verify that the fuselage can withstand some of the loads that will be applied to it during its operation, the fuselage can be assimilated to a cylindrical pressure vessel in order to apply some basic stress calculations to layer 1, which will be the one responsible for the structure's integrity, where the mechanical properties of T-680/35 Graphite/Epoxy composite material are outlined in table 14. Since the LUM will

Table 14: Physical-mechanical properties of T-680/35 Graphite/Epoxy composite material

Property	Value
Density	1600 kg/m ³
Tensile strength	2068 MPa
Elastic modulus	152 GPa
Compressive strength	1742 MPa

be responsible for resisting most of the bending and shear loads on the vehicle, the UPM will have to bear, most notably, the internal atmospheric pressure of the cabin. Internal pressures are responsible for inducing axial and hoop stress, both which work against the material's tensile strength. By applying the Von-Mises

⁸As the higher the coefficient the bigger the expansion

criterion to both stresses, the equivalent combined stress corresponds to the following equation⁹.

$$\sigma_f = \frac{\sqrt{3} Pr}{2t} \quad (12)$$

The thickness necessary for withstanding the internal pressures of a cylindrical cabin of 4m in diameter with a safety factor of 1.5 is of 0.13 mm, much lower than the 0.25 cm per layer that are proposed when following the recommendations given in reference [3]. Further analysis could be performed with FEM to verify that such a thickness is adequate in more developed design stages, but this value presents itself as a good starting point for ensuring crew safety.

Layer 4: Elastomer

The thermal mismatch produced by the difference of the thermal coefficients of layers 4 and 5 needs to be accounted for. If we didn't correct it, the layers could crack or separate from themselves. As a result, putting an elastomer layer with rubber-like characteristics will help avoid these thermal expansion disasters between the MLI and the Graphite/Epoxy layer. The candidate proposed in [3] is isobutylene isoprene.

Layer 5: MLI

Although dealing with a big temperature range is hard, thermal cycling is also a big problem over a long period of time. It can cause microdamage, making the Graphite/Epoxy less stiff and less stable. The solution proposed in [3] is to place a MLI on top of the outer Graphite/Epoxy layer (Layer 3). Additionally, this layer protects the 2 Graphite/Epoxy layers from solar radiation. If Layers 1 and 3 were to be heavily exposed to solar radiation, their properties would change and their transverse strength and modulus of elasticity would be reduced.

Layers 6 and 7

These 2 layers and the outer Graphite/Epoxy are responsible for the micrometeoroids impacts, and they will act as a bumper shield system. When a micrometeoroid impacts the bumper shield (Layer 7) it disintegrates and it's dispersed as it travels through the spacing (Layer 6) until it finally hits the back wall (Layer 3) and the remaining material is stopped. We consider the following formula [3]:

$$t_b = \frac{Cmv}{2S} \quad (13)$$

where $t_b = 0.25$ cm is the back wall (Layer 3) thickness, $m = 0.001$ g is the micrometeoroid mass, $C = 41.5 \pm 14.0$, $v \approx 13$ kms⁻¹[5] is the micrometeoroid velocity and S is the spacing [27]. If a factor of safety of 2 is used, it can be found that $S \approx 4$ cm.

Additionally, as the frequency of impacts on the Moon due to micrometeorites is really high (around 56 new craters on the Moon per year [5]), it's of paramount importance to be able to have a modular and changeable outer shell for the LEV. Aluminium will be used for the bumper shield and it will have a thickness of 1 mm, taking into account that the average diameter of a micrometeoroid impacting on the lunar surface is 1 mm[27]. As a result, different aluminium panels will be implemented as the outer shell of the LEV.

3.1.8 Upper Substitution Mechanism (USM)

In order to take advantage of the modularity of the LEV, a mechanism that will exchange different upper modules (each with a different purpose) is needed. The process the USM will perform is explained on section 4.1.3, but essentially the UPM will be moved onto a standby module while its associated LUM drives autonomously to another standby module to dock with a different UPM. See Figure 9 to understand the setup. These are the components of the USM and components on the LEV that will be relevant to this mechanism:

1. Electromagnets

Solenoids with a ferromagnetic core will be installed on the sides of both the UPM and LUM. This way their magnetic field strength can be controlled by changing the current that flows through the solenoid, and the USM can be lifted to a desired height. The magnetic field of a solenoid follows the following equation:

$$B = \mu NI \quad (14)$$

where B is the magnetic field generated by the solenoid, μ is the permeability of the ferromagnetic material used, N is the number of turns per meter and I is the intensity that flows through the wire.

⁹Where P is the internal pressure, r the radius, and t the wall thickness

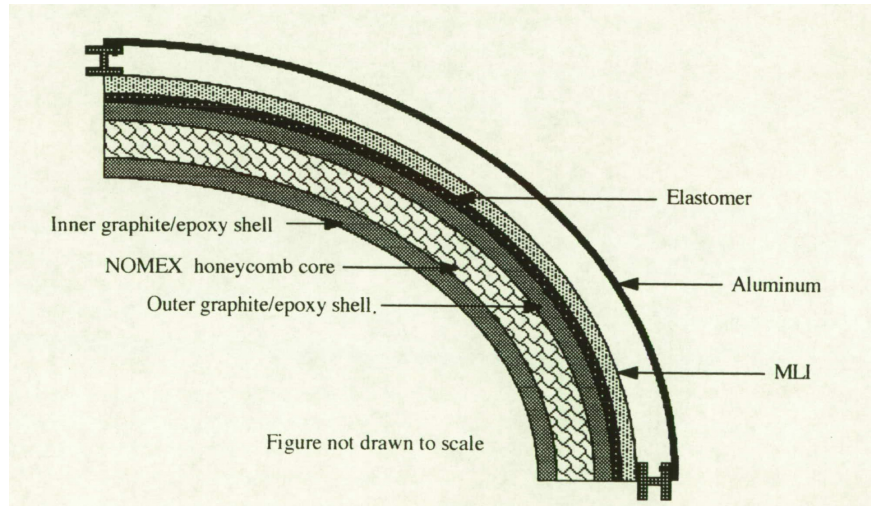


Figure 8: Hull cross section. Source: [3]

These electromagnets will be used to lift the UPM from the LUM. They need to be on both modules and with different polarity as we want a repulsive force between them. The following approximation can be used to give an idea of how big these electromagnets have to be:

$$F = \frac{\mu N^2 I^2 A}{2\pi d^2} \quad (15)$$

where N are the number of turns per meter, I the current flowing through the wire, A the total cross section of the solenoids and d the distance between the two electromagnets. If the following electromagnet data is used, just how much cross-section area A the electromagnet needs to lift the upper module d meters is obtained:

- $m = 4000 \text{ kg}$
- $g = 1.62 \text{ ms}^{-1}$
- $\mu = 6.3 \cdot 10^{-3} \text{ Hm}^{-1}$ (permeability for 99.8% pure Fe)
- $N = 1000 \text{ m}^{-1}$
- $I = 10 \text{ A}^{10}$

$$A(d) = 0.0646d^2 \quad [\text{m}^2] \quad (16)$$

This result just shows that if we wanted to lift the UPM a distance of $d = 10 \text{ cm}$, we would just need $A = 6.46 \text{ cm}^2$ in total.

2. Mechanical latches

There will be 3 on both sides of the LUM, and will be used to keep both modules together and secured.

3. Guidance and alignment system

Rails on both sides of the LUM where the UPM will be sliding from being on top of the LUM to the SLM and vice versa.

4. Attachment hooks

Located on the front and rear end of the UPM, where the winch cable will be attached.

5. Retractable mechanical arm

Responsible for attaching the cable of the winch onto the attachment hooks on the UPM.

6. Winch system

Used for pulling the UPM from the LUM to the SLM and also for holding back the upper module while the linear actuators push it from the SLM to the LUM

¹⁰The last 2 values were chosen without loss of generality after looking at common values used in industry and also trying to avoid thermal problems

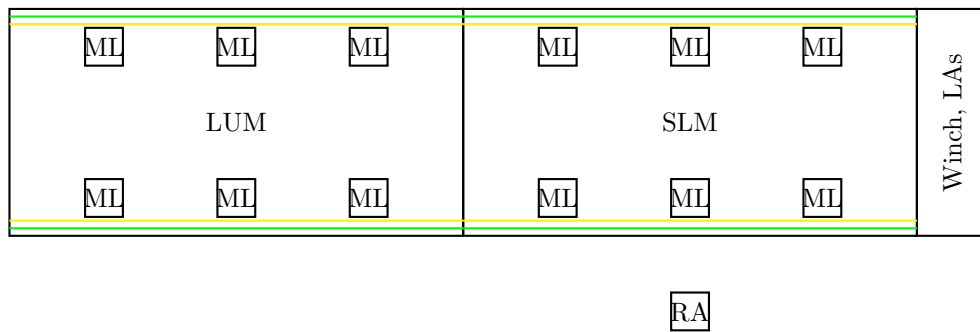


Figure 9: Conceptual diagram of the USM. ML = Mechanical Latches, LA = Linear Actuator, RA = Robotic Arm, Green Line = Rails, Yellow line = Potential area for electromagnets

7. Linear actuators

4 of them will be installed at the end of the SLM to push the UPM to the LUM.

8. Standby Lower Module (SLM)

This module will be used to place upper modules on top of it when they are not being used. Each UPM should have its own SLM, and whenever one specific upper module wants to be used the LUM will drive autonomously to the specific station (meaning UPM + SLM) and the USM will slide the desired UPM from the SLM to the LUM. Note that this module has a rail system too and the same electromagnets as the LUM to facilitate a smooth transmission. Additionally, it also has mechanical latches to secure the UPM on top of it when it's not being used in a mission.

3.1.9 Future Direction

The next step is obviously taking advantage of the modularity of this vehicle to introduce other UPMs for different purposes. The first one that comes to mind is a scientific upper module. Not only would it have a built-in state of the art lab, but it would also have a depressurization cabin to be able to perform EVAs to collect moon rocks and perform experiments on the lunar surface. Another idea could also involve a construction oriented upper module where there is a small pressurized cabin (for the operator, if needed) and different cranes and robotic arms. Also, a different UPM could be designed only for transportation of materials and devices, making the whole upper module a big pressurized cabin. As you can see, this is the main idea behind the modularity of the proposed vehicles. All of them would use the same LUM but would have different upper modules with different purposes and they could be changed using the USM.

Additionally, the idea of extrapolating these vehicles to the martian surface is not far-fetched at all. Surely, the conditions in Mars are different and almost all systems should have to be revised, but it wouldn't have to be done from scratch. Logically, Mars is the next frontier and the Moon is just seen as a stepping stone. As a result, we can expect for these vehicles to be extrapolated and used on the martian surface in the future. On another note, some of the vehicles mentioned in the previous paragraph could also be used on Earth.

3.2 Assumptions and Constraints

This paper is underpinned by a number of assumptions and constraints, discussed below:

3.2.1 Assumptions

The LEV is devised for 2040, although more realistically for the 2040s. By then, the prices to put 1 kg in space will have dramatically dropped and spaceflight safety will have improved thanks to the amount of flights performed to this date. Initially, in the 2030s, the South pole lunar base only had scientific purposes, but as it expanded several private companies saw profit either for mineral mining or space tourism and decided to invest in a for-everyone open to all public lunar base. Whether this expansion would be next to the first initial scientific settlement or not is an unresolved matter, although it could make sense to have different bases at different points to take advantage of different base selection criteria. At the same time, you could also argue that it would make sense to have this commercial expansion near the scientific base for easiness purposes.

In the 2050s a residential revolution would take place, with people being interested in buying properties on the Moon. This is the logical next step, although the authors admit that it's uncertain as it depends on the future space treaties and how they will address the owning of land on other planetary bodies. In the case that this does happen, we could be talking of different lunar cities across the lunar South pole connected with roads and a thriving lunar economy.

3.2.2 Constraints

One of the main constraints that should always be taken into account is cost. In our case, developing a large lunar pressurized vehicle is definitely not going to be cheap. Not only for the manufacturing, but there will also be costs associated with getting the vehicle to the Moon aboard a rocket (most probably Starship) and costs associated with maintenance of the vehicle.

Additionally, the generalized fear of going to space is another factor to consider. As of 2024, not many humans have been in space and as a result it is a fair judgment for people to doubt if going to space is completely safe. Consequently, the popularity and number of lunar tourists in the beginning might be low. However, as time passes by and more lunar trips have been made successfully, this perception will begin to shift and, with it, a bigger inflow of interested customers.

4 Operational Scenarios

4.1 Normal Operational Scenarios

4.1.1 Base location

As detailed in section 3.2.1, we have assumed that there is a base on the Moon big enough to include different areas with different purposes, such as: science, tourism, mining and others. The possible location of this base is an important detail, as it must have a specific set of characteristics in order to ensure the survival of the humans living in it. Additionally, the LEV is designed to take tourists in exciting but safe rides, so understanding where this base would be located is of paramount importance when designing the LEV.

The base could be located anywhere on the Moon. However, the poles are always the places considered for a permanent human base in the literature, as well as the proposed scientific missions. There are several factors that have a say in this conclusion:

- There are areas near the South pole where there is permanent sunlight throughout the year, specially on the rims of the craters. This is important for solar power production and temperature regulation.
- There are also PSRs inside the craters where trapped water ice could be found, which might be critical for water, oxygen and rocket fuel production.
- It is a strategic location, as it makes communications with Earth easier. A direct link can be established. At the same time, Earth will always appear at the same location if observed from the South pole.
- It contains a diverse and interesting terrain. It has varied rock formations that can be of scientific and exploratory interest.

The selection criteria that should be taken into account include abundance of water ice, slope of terrain, usable energy sources, communications, and base expandability among others. Note that water ice is of critical importance as it is needed for rocket fuel production and basic human survival. Although it must be confirmed with in-situ exploration, higher quantities of ice are expected in the **Sverdrup-Henson crater** than in others like *de Gerlache* and *Shackleton* [28] as seen in Figure 10. At the same time, this selected crater has a close enough flat terrain and is fairly accessible from nearby inter-crater plains.

PSR accessibility is important because they are cold enough to be hiding volatiles including ice and other minerals. The PSRs at the lunar south pole are expected to contain approximately 2.9×10^{12} kg of water ice [28], which is enough to support an initial human base taking into account that water needed for an astronaut might vary from 1 l to 3.5 l per day. Other important materials for ISRU and construction are iron, titanium oxides, nickel, plagioclase, pyroxene, ilmenite, helium-3, rare earth elements, etc.

Giovanni Leone suggests a 4.5×4.5 km base on the north eastern side of *Sverdrup-Henson* crater between the PSR and partially illuminated zone in [28]. Note that there are not permanently illuminated areas on the South Pole, but some reach 80% of the lunar day. To avoid using nuclear power and have less problems with temperature regulation, a minimum of 50 % is needed. *Henson* and *de Gerlache* share a rim that has more

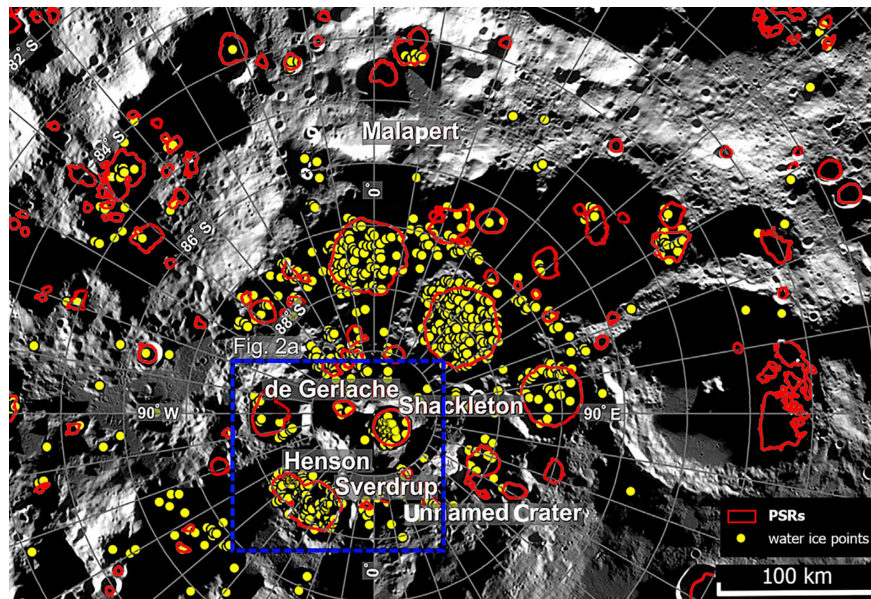


Figure 10: Water ice points at the Lunar south pole. Source: [28]

or less 80%, although nearby craters' rim also possess really good illumination conditions, optimal for solar array placements.

To choose between craters, the *Sverdrup-Henson* crater was the one with most abundance of water ice and iron detections. In terms of slope, it's consider navigable/accessible if it's below 15 degrees. *Sverdrup's* floor is really flat, generally with slopes below 5 degrees.

Lastly, the only con of setting up shop in *Sverdrup-Henson* is the fact that it only has a 40-60 % visibility of Earth, meaning that it's not the perfect place to have the solar panel arrays or for communicaitons. However, nearby massifs or the rims of *de Gerlache* and *Shackleton* craters can be exploited in order to place the solar arrays and deal with communications.

4.1.2 Launch and transportation of the LEV to the lunar surface

The LEV will travel to the lunar surface thanks to the Starship HLS rocket being currently developed for the Artemis program. This vehicle is designed to carry up to 100 tonnes of cargo to the lunar surface with payloads that can fit inside a 9m diameter cylindrical section. The rovers will be placed inside the payload bay of the lander in a vertical configuration, without any crew onboard.

The Starship rocket's flight timeline on its flight to the Moon consists of an initial launch with the HLS that will stay in orbit. There, it will wait for other variants of Starship to refuel the HLS spacecraft. Once it has been fully refueled, it will perform a TLI and park in an NRHO trajectory around the Moon. Once in the adequate orbit and time window, it will perform a de-orbit burn followed by a landing manoeuvre that will place it gently on the Moon's surface near the location of the target lunar base.

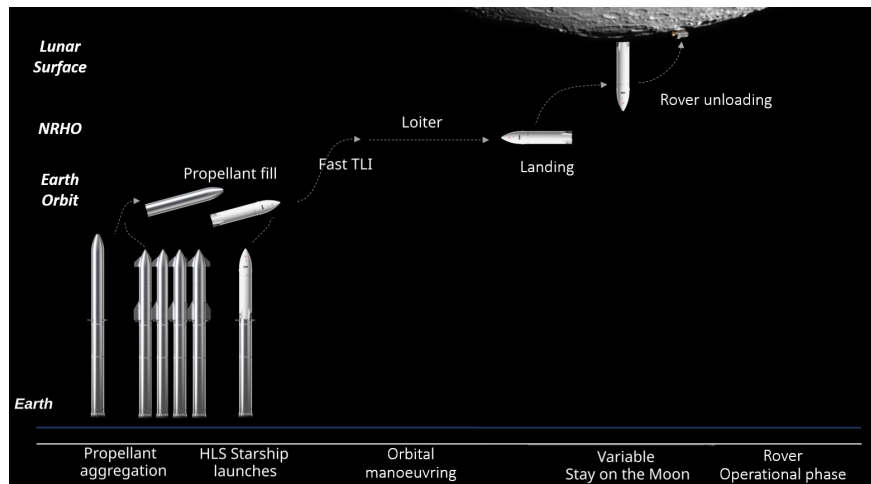


Figure 11: Diagram of the flight phases of the LEV on Starship HLS to the Moon's surface.

Once in a stable landing situation on the Moon's surface, the rover will be transported to the surface using a crane system as seen in figure 12. After this deployment has been completed, the rover will be driven autonomously towards the lunar base, where it will dock, and its normal operational phase will begin.

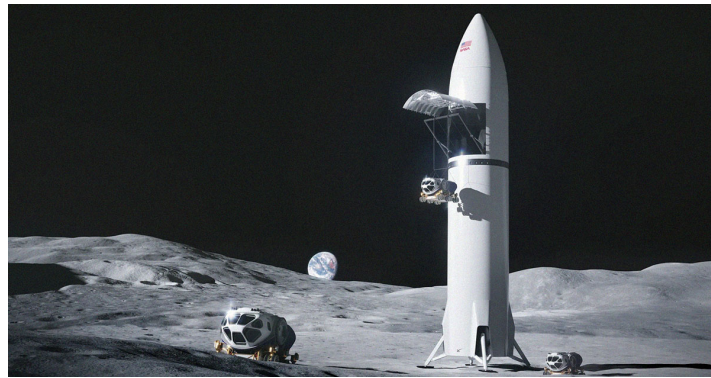


Figure 12: Render of the Starship HLS lander unloading rovers to the lunar surface.

4.1.3 Changing UPMs with the USM

The following process explains how to move the UPM from the LUM to the SLM. After the upper module is secured and in standby, the lower module can drive autonomously to a different SLM with a different upper module stationed on top of it.

1. Position the LEV next to the SLM so that the UPM can move in a straight line.
2. Deactivate mechanical latches on the LUM.
3. Attach the winch's cable with the hook on the UPM using the mechanical robotic arm.
4. Lift the UPM using the electromagnets by increasing opposing currents.
5. Pull UPM after activating the winch and guide it with the rails.
6. Stop winch and disengage electromagnets whenever the UPM is in standby position.
7. Secure UPM on top of the SLM by activating mechanical latches.

Next, the detailed process by which a UPM will be pushed from its SLM to a LUM is outlined:

1. Position the LUM next to the SLM so that the UPM can move in a straight line
2. Deactivate mechanical latches on the SLM
3. Attach the winch's cable with the hook on the UPM using the mechanical robotic arm
4. Lift the UPM using the electromagnets by increasing opposing currents
5. Push the UPM with the linear actuators and control the speed using the winch system

6. Turn off all electromagnets when the UPM is properly stationed on top of the LUM
7. Activate mechanical latches on the LUM

4.2 Maintenance Scenarios

Several maintenance scenarios will take place along the lifespan of the rover with the aim of assuring its correct continued operation. A list of the common maintenance operations to be performed on the LEV is given in this section.

4.2.1 Fuel cell refueling and bi-product removal

The fuel cell has a limited fuel supply that dictates the distance that the rover can traverse. The hydrogen and oxygen fuel tanks will have to be replenished to avoid running out of energy. Additionally, the oxidation reaction that takes place inside the cell leaves behind water, which must also be emptied before the water tank is full.

A draining port will be installed on the LUM to extract the produced water and use it back on the lunar base. Additionally, two ports specifically made for pumping the fuel gases will be used for filling the fuel containers and pressurizing them up to correct working conditions. This process can be performed manually (likely to be done in the first pathfinder missions), but can be automated in later stages of the rover operations using robotic arms or similar machinery.

4.2.2 ECLSS replenishment and maintenance

The ECLSS is a complex subsystem that requires regular maintenance operations from mission to mission. These operations are critical, as the well being of the crew directly depends on this subsystem working nominally.

One step in the ECLSS maintenance process consists of replenishing the oxygen and nitrogen storage tanks, which are used for various reasons such as regular metabolic consumption, as well as for countering pressure leaks in the cabin, or feeding air to the pressure suits in the case of a more severe leak. Access ports to these tanks are placed in the UPM with easy access to either maintenance crew or automatized maintenance machinery from the module's exterior.

The air revitalization section of the ECLSS also needs to be treated every 2-3 missions by replacing one of the Lithium hydroxide (LiOH) cartridges used for scrubbing the waste CO₂ resulting from the crew breathing. The contaminant particle filters also have to be replaced with the same periodicity. This operation has to be performed from the inside of UPM itself by the maintenance crew under conditions of equalized pressure with the rest of the lunar base.

The last step of this maintenance process consist of removing the other forms of accumulated human waste that the Life Support segment of this subsystem collects, mostly human feces and urine. External ports on the UPM are also provided for easy access to the waste tanks. These residues will then be pumped to the lunar station for proper treatment, such as recycling the urine back to usable water, or using then for agricultural purposes.

4.2.3 UPM technical inspection

The proper condition of the technical subsystems in the interior of the UPM such as the ECLSS, the OBC, the backup power source, the electrical connections, or the user interfaces for operating the vehicle (among others) will have to undergo a strict technical inspection that ensures that the equipment can continue operating without any expected failures.

In case that any issue is detected in the inspection process, the specific UPM will be grounded, and an investigation process will be carried out to determine the possible cause of the problem and how to fix it. Once the issue has been solved, a second inspection process will be put in place to make sure that the fix itself works properly together with the rest of subsystems and, if successfully passed, the previously faulty UPM will be put back into service.

If the second inspection is not passed, the investigation process will not be closed. If the damages to the module are considered to be beyond repair, the specific UPM will be dismantled, and the parts that can be re-used for future repairs will be catalogued and properly stored.

4.2.4 LUM technical inspection

The various LUM platforms have to be inspected before starting any mission to ensure that the vehicle is fully capable of correct mobility on the lunar's surface, as well as of correct autonomous or semi-autonomous navigation, and of communication with the UPM and with the ground station.

The first aspect to check regarding the rover's mobility is its structural integrity. A mechanical inspection has to be performed to ensure that all of the suspension elements are correctly attached to the rest of the body, and that the chassis is holding up without any mechanical failures that could be induced by fatigue or some previous excessive load. This can be performed via a visual inspection of the critical points of the LUM, which could potentially be automatized using computer vision for fault detection.

A system-level check also has to be performed on the electrical assemblies, making sure that the fuel cell is able to properly deliver the power to the motor drivers and the other subsystems. The correct functioning of the electronics of the vehicle can also be easily performed. This step could be automatized using several sensors placed in critical locations of the module.

If any specific failure is detected during the inspection process, the same protocol as the one outlined in section 4.2.3 will be followed.

4.2.5 Replacement of wheels due to wear

A specific maintenance scenario that stems from the more general case outlined in section 4.2.4 describes the case where a wheel (referring to the NiTiNOL mesh) has been either damaged or worn out, either from continued use or from an encounter with a particularly damaging obstacle while driving. In this case, the wheel has to be replaced.

The main difficulty with this procedure lies in the fact that the spring-mesh tire assembly itself has to be changed independently from the wheel's hub or rim, as the motors and other critical subsystems are housed inside it. This, however, is already taken into account in the wheel's design, and the spring-tire assembly includes a ring frame that keeps the wheel's adequate shape and tension despite being removed from the complete wheel assembly. This may need additional specialised tools, which will be included in any rescue missions to replace the wheel.

If the tire is replaced in a preventive manner due to mechanical wear, only the NiTiNOL mesh will have to be replaced, allowing for the ring-frame to be reused. This will require specialised workers performing manual labour, as achieving the right tension and wheel shape could be a complicated task without overly specialised tooling, but it will allow for more parts of the rover to be reused.

4.2.6 Interior cleaning

The interior of the vehicle will have to be properly cleaned and sanitized after each mission to ensure an immaculate interior and a positive experience for the customers. This will also serve to collect other forms of waste that have accumulated, such as food packaging. A specialized cleaning crew will be in charge of this process.

4.2.7 Replacement of the emergency battery

In the case that the power that is being generated in the LUM either stops reaching the UPM or stops being produced altogether, the secondary battery will be used to power the stand-by systems, as described in section 4.3.7. The back-up battery, however, is non-rechargeable and will have to be replaced with a new unit, after the old one has been used.

The replacement of this device will have to be performed manually, as the backup battery sits on the inside of the module, and it can be performed using basic tools without needing specialised crew, once docked to the lunar base.

4.3 Failure Scenarios

Various failure scenarios can arise during roving operations on the Moon's surface. Such failure modes have been studied in depth so that their potential negative outcomes can be mitigated through the use of redundancy designed into the rover and emergency protocols that include rescue missions, carefully planned to ensure the safety of the crew.

4.3.1 Launch failure

If the launch of the rover vehicles experiences a failure which implies that the rover cannot reach its destination, no crew members will be in compromise as the rover vehicles will be shipped separately to the crew capsules that take the passengers to the Moon's surface. The vehicle, however, will be completely lost.

4.3.2 Landing error

In the case of a landing error that causes the rover to touch down in the wrong location of the Moon's surface, the rover's autonomous capabilities will be used to drive towards the lunar base as long as it falls within the total distance that the rover can traverse. Since any errors in landing location are expected to be small, this solution could be feasible. If the landing error is too large the rover will be lost, the crew will not onboard in this stage.

4.3.3 Door / docking system malfunction

In the case of a docking failure that prevents the transfer of crew from the rover outwards, external supplies will be fed through external oxygen-nitrogen ports such that the lifespan of the ECLSS can be lengthened. This will give the engineers on the ground enough time to identify and solve the issue with the docking system, or come up with an alternative solution for evacuating the crew. The replacement CO₂ scrubbers included on the inside of the rover (mentioned in section 4.3.8) will also be used for expanding lifetime of the ECLSS.

In the rare case that an issue with the docking hatch causes it to remain shut, and this cannot be fixed within the extended ECLSS lifetime, the UPM will be placed inside a pressurized workshop, large enough to house it, where destructive methods will be used for evacuating the crew. The pressurized workshop will be necessary for not exposing the crew to the full vacuum of the Moon, despite having their pressure suits. The UPM which has been forced to open destructively will have to be discarded from further use, but some of its subsystems may be stored for future repairs if they remain in good condition.

4.3.4 Mechanical failure of the tractive system

If the tractive system of the vehicle, which includes the wheel (tyres, motors and breaks sub-assemblies), the suspension elements, or the steering system experiences a sudden failure such that the mobility of the vehicle is hindered or completely halted, a rescue mission will be sent out towards the stranded vehicle.

During the time in which the rescue mission arrives, the stranded crew will attempt to determine the exact cause of the failure. If this is impossible to do from the vehicle's inside, the rescue crew will have to determine this upon arrival, possibly necessitating an EVA mission. After the failure cause is identified, the rescue team will attempt to repair the vehicle, as long as doing so is possible in a low enough time span. If this is not possible, a crew transfer rescue mission as outlined in section 4.3.12 will be carried out, and the stranded vehicle will be abandoned. Once the crew is already safe, a second rescue mission for the rover alone could be proposed if deemed feasible.

4.3.5 Micrometeoroid impact

In the case of non-predicted micrometeoroid impacts, their size is statistically expected to be absorbed by the meteoroid protecting layers incorporated in the rover's walls and windows without causing disruptions to the rover's operation. Depending on the severity of the impact, however, an emergency protocol may be put into place.

If a micrometeoroid manages to pierce through or crack a wall or window, the emergency protocols outlined in section 4.3.11 will be put into place. If the impact instead affects a critical and less-protected part of the rover such as the radiators the protocol outlined in 4.3.10 will be put into place. If the wheels or the motor/breaking assembly are impacted and damaged beyond repair, the emergency protocol outlined in section 4.3.4 will be employed.

4.3.6 Onboard Computer failure

The OBC is a very complex system which could fail in various ways. Additionally, the OBC is divided into several compute units, two in the LUM with one providing low-level control of the vehicle and the other one providing high-level autonomous capabilities, and a third computer unit in the UPM in charge of this module's subsystems, as well as the user experience and interfacing between the two modules.

All of these components are critical to the correct functioning of the vehicle, yet have a large number of modes of failure, from faulty code to radiation upset events. To avoid the faulty code from causing issues, good programming practices will be enforced alongside a continuous integration workflow that tests the code in an automatized and measurable manner, such that the edge cases of the computers can be assessed and no critical errors get embedded into the final code-base. Despite this, if they compute units encounter any problem that impedes their correct functioning, reboot sequences automated via an internal watchdog timer shall begin in order to get the OBC up and running again.

The other possible radiation related issues such as any SEU or Single Event Upset (commonly known as bit-flips) are a common occurrence in spaceflight. These will be fought against through radiation-hardening of critical components, together with redundant computing techniques like TMR (where three independent microprocessors perform the same computations and then voting logic is used to detect an outlier in the results) and ECC memory, which can detect and correct data corruption in the computer's memory.

4.3.7 Power system failure

In the case that the main power hydrogen-oxygen fuel cell that is housed on the LUM fails to successfully output electrical power, two scenarios can occur.

If the power output is decreased but not completely cut, the exact amount of power that is left available will determine if a return drive is feasible, possibly at lower speeds than nominal, where the power needed for the UPM could be supplemented by the backup battery if needed.

If the power available indicates that there will not be enough energy required for a successful return mission (a decision that can be taken with the help of the OBC) or if the power from the fuel cell is cut entirely, a rescue mission can be initiated as outlined in section 4.3.12. In this case, the backup battery would start powering only the subsystems that are necessary for the crew's survival and communication with the rescue team.

If the failure is caused by a leak of either oxygen or hydrogen, special care has to be taken to avoid the gases from igniting and/or detonating, especially if both gases become mixed. For this reason, the hydrogen-oxygen tanks will be shrouded by separate containers with pressure relief paths for the gases to safely escape the exterior, avoiding any potentially dangerous fuel concentrations outside the tanks.

4.3.8 Carbon dioxide removal system failure

In the case that the CO₂ scrubbers from the air revitalization section of the ECLSS fail, additional scrubber modules are included inside the cabin, which can be manually replaced without complexity. This would allow the crew to return back to the lunar base so that an investigation of the failure can be carried out.

4.3.9 Oxygen delivery system failure

In the case that the ECLSS oxygen supply fails to deliver its contents to the cabin (an unlikely event due to the double redundancy in the feed mechanisms), compact chemical emergency generators, commonly known as oxygen candles, are included. These can be activated to produce oxygen for the crew to breath at the same rate of their average oxygen consumption, allowing for a drive back to the lunar base for further investigation.

4.3.10 Thermal control system failure

In the case that a failure is detected in the thermal control system, different courses of action can be taken, depending on the origin of the problem, its magnitude and its cause.

If the drivetrain begins overheating due to a leak of coolant fluid, the rate of the leak will determine the procedures to take. If the leakage rate is low enough, a drive back to the lunar base will be attempted. If it is too high to drive back under normal conditions, a return drive at very low speeds will be considered (this could be accomplished by splitting the power loads between the main and backup power sources in order to accommodate the longer mission duration), where the ECLSS lifetime as well as the storage of crew consumables will have to be considered as a determining factor when deciding the course of action. If a return drive is by no means possible, a rescue mission will have to be carried out, as outlined in section 4.3.12.

Another possibility is that the passive cooling mechanisms such as the radiators become ineffective. This could be caused, for example, due to an excessive deposition of lunar dust due to a build-up of static

electricity. In this case, the issue could be addressed by first trying to clean the radiators, or dimming the PDL windows to cool the interior of the vehicle, and re-assessing the heat dissipation capabilities. Then, the a protocol similar as outlined for the case of a coolant fluid leak but for the upper module's thermal management shall be considered. Additionally, a low-power mode could be entered, where all subsystems and power-dissipating devices would be shut-down except those strictly necessary for the return to base, lowering the heat loads.

Lastly, the opposite issue could occur, where the heating elements of the climate control system or the thermal conditioning systems for the electronics could fail. These systems, however, include double redundancy and this scenario can be avoided within tolerable risk.

4.3.11 Cabin air pressure leak

If an air leak from the cabin occurs which is minor and easily accessible, it can be identified with the Acoustic Imaging Cameras for Pressurized Leak Detection that the rovers will be equipped with, and the included on-board patching equipment will be used to seal the leak spot in the cabin. If the hole is not easily accessible, however, additional oxygen and nitrogen is included in the tanks of the ECLSS such that additional air can be pumped to the cabin while the rover initiates a return drive to the lunar base.

If the air leak is worrisome enough, the visor of the pressure suits that the crew will be carrying can be automatically closed and, following the correct emergency procedures, the ECLSS quickly switch to maintain the correct conditions inside the individual pressure suits of the astronauts.

4.3.12 Crew transfer rescue mission

In this section, the worst-case scenario that has been found in the case of a rescue mission is analyzed. Here, the rover travels out to its maximum operating range, stays still in stand-by mode for its maximum design time (corresponding to three hours) and, before initializing the return trip, a mishap occurs that impedes its mobility.

After confirming that the rover is incapable of traveling back to base, a rescue mission is sent which, once it has reached the stranded rover, has to perform some crucial rescue tasks. These can occur under three main scenarios:

- In the case that the rover's mobility is impeded due to an external obstacle, the rescue rover could force the stranded rover clear of its obstacle by towing it. While this operation will consume additional energy in the towing process, it is bound to require less time than a crew transfer.
- If the rover's inability to move comes from a greater internal cause that cannot be fixed on the spot, one solution could mean towing the stranded rover back to the lunar base. While this could be accomplished in some scenarios, the mission would be likely to require large amounts of excess energy to be spent, as well as a much greater total mission time. In such a case, the priority would be returning the crew back to base, even if it means the loss of a rover (until a posterior rescue mission is sent, this time with much lower stakes as the crew will already be in a safety situation).
- The most feasible rescue mission would involve a crew transfer from the stranded rover to the rescue rover, for it could be performed even if the rover cannot be liberated from its external obstacles, and it would prioritize the crew's well being while being energy and time efficient. This is the case that is going to be studied most extensively and presented as the main rescue scenario.

Considering the crew transfer rescue operation in more detail, it can be broken down into the following tasks.

1. Approximation and alignment
2. Soft capture
3. Hard docking
4. Crew transfer
5. Undocking
6. Return to base

The complete rescue mission's timeline has been laid out using calculations of the driving time as well as mission analysis of existing docking procedures. The detailed timeline is presented in table 15, where A stands for the stranded rover and B stands for the rescue rover.

Table 15: Crew transfer rescue mission timeline.

Sub-task	Region	Rover	Start time (h)	End time (h)	Delta time (h)
Initial drive	Lunar terrain	A	0:00	5:00	5:00
Standby time		A	5:00	8:00	3:00
Gets stuck	Stranded site	A	8:00	8:30	0:30
Emergency notice		A	8:30	8:35	0:05
Response time	Lunar base	B	8:35	9:05	0:30
Hatch closing		B	9:05	9:10	0:05
Safety checks		B	9:10	10:45	1:35
Undocking		B	10:45	10:50	0:05
Departure		B	10:50	10:52	0:02
Rescue drive	Lunar terrain	B	10:52	15:52	5:00
Approach	Stranded vehicle	B	15:52	16:07	0:15
Alignment		B	16:07	16:22	0:15
Soft capture		A-B	16:22	16:28	0:06
Hard docking		A-B	16:28	16:33	0:05
Umbilical deploy		A-B	16:33	16:35	0:02
Pressure suit removal		A-B	16:35	16:37	0:02
Safety checks		A-B	16:37	18:12	1:35
Hatch opening		A-B	18:12	18:17	0:05
Crew transfer		A-B	18:17	18:47	0:30
Hatch closing		A-B	18:47	18:52	0:05
Pressure suit dress-up		A-B	18:52	18:58	0:06
Safety checks		B	18:58	20:33	1:35
Undocking		B	20:33	20:38	0:05
Departure		B	20:38	20:40	0:02
Return drive	Lunar terrain	B	20:40	25:40	5:00
Approach	Lunar base	B	25:40	25:55	0:15
Alignment		B	25:55	26:10	0:15
Soft capture		B	26:10	26:16	0:06
Hard docking		B	26:16	26:21	0:05
Umbilical deploy		B	26:21	26:23	0:02
Pressure suit removal		B	26:23	26:25	0:02
Safety checks		B	26:25	28:00	1:35
Hatch opening		B	28:00	28:05	0:05

Using the results extracted from table 15, the total operational time that each rover must endure in the most critical failure scenario can be obtained, as is presented in table 16.

Table 16: Crew transfer rescue mission timeline breakdown.

Mission portion	Elapsed time (h)
Total mission time	28:05
Driving time A	5:30
Stand-by time A	13:28
Driving time B	10:34
Stand-by time B	6:43

5 Summary of Impacts

The introduction of a pressurized lunar vehicle designed for space tourists would indeed provoke a paradigm shift in the space industry. In the beginning it may only open the space economy for rich people, but as time goes this idea will push for space to be available for everyone. The lunar economy will thrive and it will be the first step for our journey to Mars. Additionally, developing all the components of the LEV will require improving current technologies and developing new ones.

Although sending vehicles with a mass of 6000kg with SpaceX's Starship would not be the most environmentally-friendly activity you could hope for, the fact that the LEV is modular should be taken into account. Let's say we want to launch 6 different LEVs, that would involve using the combustibles necessary to bring at least 36000 kg to the lunar surface. However, since the LEV is modular, an arrangement could be made where only 2 LUMs were stored inside Starship (and obviously the 6 different UPMs). Therefore, this would decrease the total mass to 20 000 kg¹¹ and the combustibles needed for it. This reduction of mass will not only have a positive impact on the environment but it will also be a cheaper alternative, and all thanks to the LEV's modularity.

6 Conclusion

A lunar pressurized vehicle has been described as an off-road vehicle capable of carrying 6 space tourists through the unforgiving lunar terrain for a maximum duration of 10 hours with a weight of 6 tonnes and dimensions of $4 \times 4 \times 8$ m, highlighting its modularity and human-oriented design. The modularity of the vehicle will allow different upper modules with different mission profiles to be paired with the same lower module, reducing costs and improving efficiency. The lower module will normally include the wheel tracks and other necessary machinery and tanks, while the upper modules will normally contain the pressurized cabins and the ECLSS.

The other important factor in vehicle design has been the willingness to make the LEV comfortable for space explorers. Interiors have been optimized to give extra space to space travelers and big windows have been prioritized in order to provide higher visibility. Although not specified in this document, the interior was assigned a dynamic layout to change seats into different settings, allowing to have different configurations for different purposes. For example, the seats can be moved next to the windows for seated observation of the Moon or they can be moved to the center of the vehicle to allow space tourists to walk.

Although the lunar economy is still growing, it has the potential to become an important part of the space economy in less than 15 years. Designing the LEV has been highly insightful for the understanding of the limitless possibilities that a lunar South Pole base would provide for humankind. Therefore, at IBO Engineering we are optimistic about the LEV but we are also aware that an imminent commercialization is unfeasible. It should be understood, for now, as an exercise to showcase IBO Engineering's ability to design vehicles for different environments and deal with potential obstacles that space travelers will face in the following decades.

¹¹LUM's mass is 4000 kg and UPM's mass is 2000 kg

7 Exhibits

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